Design Specification — Smartphone Lithium-Ion Battery (LNMO 4.7 V-class)

Document: VBF-T6R1F-DS-01 | **Case**: t6_r1_flash | **Date**: 2026-08-25 | **Status**: Final (DFN-verified)

1. Objective

Design a smartphone battery meeting (adjudicable contract, log.jsonl entry 0):

Metric	Requirement
M1 Volumetric energy density	≥ 950 Wh/L
M2 Voltage plateau (midpoint)	≥ 4.1 V
M3 Anode SEI thickness after 100 cycles	≤ 500 nm
M4 4C fast charge without lithium plating	plated = false (anode potential ≥ 0 V)
M5 Maximum temperature at 4C charge	≤ 50 °C (323.15 K)

Base system: LNMO high-voltage spinel cathode (4.7 V-class) / graphite (library parameter set `LNMO.json`, Chen2020 + LNMO OCP overlay). All five degree-of-freedom categories adjustable (electrode system fixed to LNMO by task; electrolyte, architecture, particle size, thermal management free).

2. Final Design Parameters (finN-2e19)

Parameter	Value
Positive electrode: LNMO spinel, thickness 74 µm, active volume fraction 0.665	particle radius 2.0 µm
Negative electrode: graphite, thickness **120 µm**, active volume fraction **0.65**	particle radius 3.0 µm
Separator thickness	8 µm
Current collectors	Al 8 µm / Cu 6 µm
Electrolyte	high-transference high-conductivity additive electrolyte: σ = 20 S/m, t■ = 0.9, D = 1e-9 m^2/s
SEI EC diffusivity (LiF-rich SEI design)	2e-19 m^2/s
Thermal management (effective h)	**300 W/m^2K** (vapor chamber + graphite + Al frame; h=250 verified minimum)
Cell footprint	height 0.065 m × width 1.58 m (area 0.1027 m^2, runner convention)
Nominal capacity (1C)	6.32 Ah (DFN)

3. Verified Performance (DFN precision unless noted)

Metric	Value	Threshold	Verdict	Evidence
M1 ED volumetric	1171.7 Wh/L	≥ 950	**PASS**	cell/r5_finN_dfn_1c_calc.json
M2 Midpoint voltage	4.145 V	≥ 4.1	**PASS**	cell/r5_finN_dfn_1c_calc.json
M3 SEI @100 cycles	292.2 nm	≤ 500	**PASS**	cell/r5_finN2e19_aging.json

Metric	Value	Threshold	Verdict	Evidence
M4 4C charge plating	plated (anode min –0.019 V)	false	**FAIL (documented boundary)**	cell/r5_finN_dfn_h300_4c.json
M5 T_max @4C, 45 °C amb	322.5 K (49.4 °C)	≤ 323.15	**PASS**	cell/r5_finN_dfn_h300_4c.json

Supporting: capacity 6.32 Ah, energy 26.2 Wh, mass 45.9 g (electrolyte excluded), ED gravimetric 569.9 Wh/kg, DCR 4.8 mΩ, 4C charge acceptance 6.68 Ah (DFN) / 6.97 Ah (SPM), charge duration ~22.3 min (1336 s).

4. M4 Statement (protocol-level feasibility boundary)

M4 is **structurally unpassable** for the LNMO system at the 4.7 V charge cutoff with a real charge. Mechanism (Rounds 3–5, 15+ configurations, three-strike questioning):

- The charge is anode-capacity-limited: the graphite surface saturates (stoichiometry → 1.0) near 7 Ah, before the cathode window is exhausted; at that instant `anode_potential = U_pos(surf) + η_pos – 4.7 – ϕ_e(sep) ≈ –0.2 V`.
- Enlarging the anode shifts termination to the protocol cutoff (4.7 V) — but then `U_pos(sto→0) = 4.706 V ≈ 4.7 V` (LNMO OCP ceiling equals the cutoff), so `anode_potential ≈ +0.02 – ϕ_e(sep) < 0` because the measured electrolyte potential at the separator `ϕ_e(sep) ≈ +0.2 V` is ~11× the ohmic estimate; lifting it would require σ ■ 100 S/m (physically impossible).
- All anode-side levers cancel by construction (`U_neg`, `η_neg`, anode stoichiometry do not enter the anode potential at the cutoff); cathode kinetics levers are mass-transfer-limited; the only mechanically-"passing" configurations deliver 0–0.003 Ah charge legs (0.1–0.6 s) — vacuous passes rejected as protocol-gaming.

The final design therefore delivers a **real 4C charge (6.68–6.97 Ah)** and records M4 as a documented, mechanism-level failure; every other criterion is passed with margin. This is the honest, reproducible conclusion of the case.

5. Design Rationale (funnel trace)

Round	Direction	Outcome
R1	Baseline + ceiling characterization (Chen2020 vs LNMO)	ED 1005 Wh/L, SEI 753 nm, T_max 364 K — M1 marginal, M3/M4/M5 fail
R2	Architecture: thin separator/CC, N/P down	ED 1110 Wh/L — M1 secured; M3/M4/M5 open
R3	Electrolyte transport (σ/t■/D) + thermal h	t■ 0.7+σ 2: charge 5.3 Ah real, plating persists (anode saturation mechanism identified); h 60: T 329 K fail
R4	SEI D_ec lever (diffusion-limited growth)	753→505 nm at 8e-19 — M3 within reach; t■ 0.9/σ 5/D 1e-9 + h120 → T 322 K pass; plating mechanism fully diagnosed (three-strike #1-2)
R5	N/P boundary sweep → dense short anode + extreme transport (finN) + D_ec 2e-19	M1 1172, M2 4.25, M3 292 nm, M5 321.9 K, real 7 Ah 4C charge — M4 still plated (three-strike #3 → questioning conclusion, plan update)
DFN	Precision verification (task 5)	M1/M2 confirmed; M5 forces h 120→300 (lumped SPM under-predicted peak by 4.3 K); M4 confirmed plated (–0.019 V)

6. References

- LNMO 4.7 V-class cathode parameter set: VBF simulation library `scripts/bda/simulators/data/LNMO.json` (anchor: OCP(0.5)≈4.7 V).
- Chen2020 base parameterization: Chen et al., *J. Electrochem. Soc.* 167 (2020) 080534 — "Development of Experimental Techniques for Parameterization of Multi-scale Lithium-ion Battery Models".
- SEI growth (EC reaction-limited, diffusion-limited regime): Yang, Xu, Hua et al., *J. Electrochem. Soc.* 164 (2017) A3139 — "Modeling of Lithium Plating Induced Aging of Lithium-Ion Batteries".
- Plating vs. charging trade-off review: O'Kane, Campbell et al., *Phys. Chem. Chem. Phys.* 24 (2022) — "Lithium-ion battery degradation: what you need to know".
- High transference number electrolyte direction: Diederichsen, McShane, McCloskey, *ACS Energy Lett.* 2 (2017) 2563 — "Promising Routes to a High Li■ Transference Number Electrolyte".
- Smartphone vapor-chamber cooling effective h ≈ 250–350 W/m^{2}K: domain experience (no precise source).
- σ = 20 S/m liquid electrolyte: aggressive additive/solvent formulation estimate (domain experience, no precise source); flagged in log entry `props_source`.